ASEAN GEOAI FUSION 2026

Universiti Kebangsaan Malaysia
The National University of Malaysia

MCMC

AGAIF 2026 BOOTCAMP 1: FOUNDATIONAL GEOAI

Introduction to GeoAI & Geospatial Session 2

Online
7 Juy 2026 (Tuesday)

Prof. Sr Ts. Gs. Dr. Khairul Nizam Abdul Maulud

Department of Civil Engineering,
Faculty of Engineering and Built Environment,
Universiti Kebangsaan Malaysia





GIS SPATIAL ANALYSIS
MODULE OUTLINE

Geospatial Skills for a Smarter ASEAN

Master key spatial analysis techniques to transform location-based data into actionable insights for sustainable development across ASEAN.

WHAT YOU WILL LEARN

- Understand core spatial analysis concepts
- Convert spatial data accurately
- Combine and analyze layers
- Create buffers and assess proximity
- Apply skills to real-world ASEAN challenges

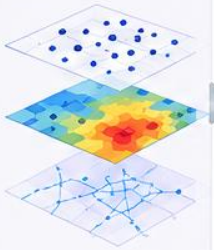
REAL-WORLD IMPACT IN ASEAN

- Stronger disaster preparedness
- Sustainable natural resource management
- Smarter urban & rural planning
- Data-driven policy & decision-making

1 SPATIAL ANALYSIS CONCEPTS

Understand how location, patterns, and relationships help us explore and explain the world.

- Identify patterns & trends
- Compare spatial relationships
- Draw conclusions & create new datasets



ASEAN APPLICATION SCENARIO

Flood risk mapping in the Philippines

Identify high-risk areas to support early warning and community safety.



2 SPATIAL DATA CONVERSION

Convert data between different coordinate systems to ensure accuracy and consistency.

- Different map projections
- Datum & coordinate systems
- Transform spatial data for seamless use



ASEAN APPLICATION SCENARIO

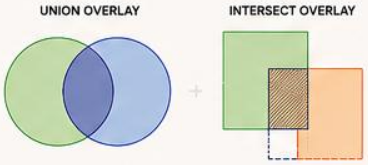
Infrastructure planning in Vietnam

Integrate datasets from multiple sources for roads, utilities, and public facilities.

3 OVERLAY OPERATIONS

Combine layers to discover relationships and make better decisions.

- Overlay analysis (Union, Intersect)
- Generate new features from overlapping data



ASEAN APPLICATION SCENARIO

Environmental management in Thailand

Overlay land use and forest cover data to support conservation planning.



4 BUFFER ANALYSIS & PROXIMITY

Create zones around features and analyze proximity to assess risk and accessibility.

- Buffer & multi-ring buffers
- Identify critical zones
- Find nearby features



ASEAN APPLICATION SCENARIO

Port development in Indonesia

Buffer around ports to assess environmental impact and plan logistics zones.



WHY IT MATTERS FOR ASEAN

- Better disaster preparedness
- Sustainable natural resource management
- Smarter urban & rural planning
- Data-driven policy & decision-making
- Stronger regional collaboration

ASEAN IN ACTION – EXAMPLES

- **Philippines** Flood risk assessment
- **Vietnam** Infrastructure mapping
- **Thailand** Land use planning
- **Indonesia** Port & logistics planning
- **Malaysia** Urban growth analysis
- **Cambodia** Watershed management

... and more across ASEAN

Accurate data. Smart analysis. Sustainable future for ASEAN.

GIS SPATIAL DATA

Models & Formats Overview

GIS uses two main data models—Vector and Raster—to represent the real world. Each model has unique characteristics and data formats.



TWO TYPES OF SPATIAL DATA MODELS

VECTOR MODEL

Represents objects as points, lines, and polygons.

Objects are created by connecting points with straight lines or arcs, and area is defined by sets of lines.

POINT LINE POLYGON



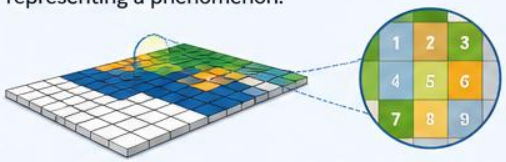
REAL WORLD

Both models aim to represent features and phenomena as they exist in reality.

RASTER MODEL

Represents the study area as a grid of equally sized cells (pixels).

Each cell contains a single value representing a phenomenon.



GIS DATA FORMATS

RASTER FORMATS

Remote Sensing Image (SPOT, Landsat, Worldview etc)	Tagged Image File Formats (TIFF)	Portable Network Graphics (PNG)
Geo Tagged Image File Formats (GeoTIFF)	Multi-resolution Seamless Image Database (MrSID)	Graphic Interchange Format (GIF)
Joint Photograph Experts Group (JPEG2000)	JPEG File Interchange Format (JFIF)	DEM, DSM, DTM
Enhanced Compressed Wavelet (ECW)	Digital raster graphic (DRG)	Shuttle Radar Topography Mission (SRTM)

VECTOR FORMATS

Hardware Specific Formats	Arc-Info Coverage
Shapefile	Arc-Info Interchange File (.e00)
Digital Exchange Format (DXF)	GeoDatabase
Digital Line Graph (DLG)	Spatial Database engine (ArcSDE)
TIGER	Esri TIN
Scalable Vector Graphics (SVG)	GeoJSON

KEY TAKEAWAY

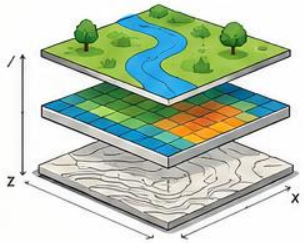
VECTOR MODEL
Best for discrete features like roads, rivers, buildings, boundaries.

RASTER MODEL
Best for continuous surfaces like elevation, temperature, imagery.

CHOOSE THE RIGHT FORMAT
Selecting the appropriate data format ensures compatibility, efficiency, and better analysis.

GIS DATA CONVERSION

Transforming Data, Enabling Better Decisions



Data conversion is the process of transforming GIS data from one format to another to ensure compatibility, interoperability, and effective analysis.

KEY POINTS

WHAT IS IT?
A mechanism for converting GIS data from one format to another.

CAN BE EASY OR DIFFICULT
It depends on the speciality and complexity of the data formats.

PROPRIETARY FORMATS
Require special translators or software tools for conversion.

COORDINATE CONVERSION
Includes converting coordinate reference systems (CRS).

EXAMPLES OF COORDINATE CONVERSION

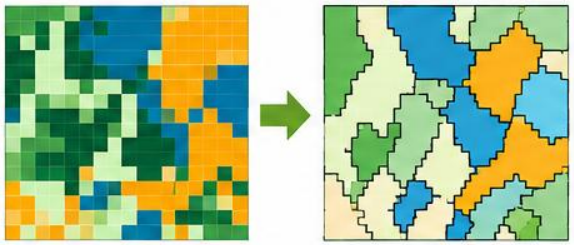
Cassini ⟷ RSO ⟷ GDM2000 ⟷ WGS84

CONVERTING RASTER DATA TO VECTOR FEATURES

01 CONVERTING RASTER DATA TO POLYGONS

- Contiguous cells with the same value are converted into polygons.
- Arcs are created from cell borders in the raster.
- NoData cells in the input raster do not become polygons in the output.

RASTER INPUT → POLYGON OUTPUT

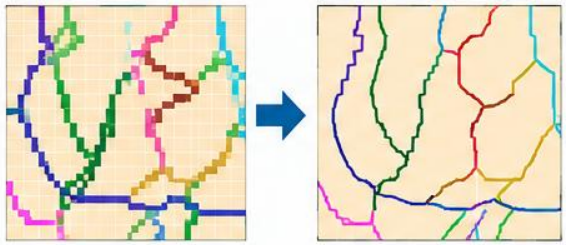


USE CASE
Land cover mapping, soil classification, administrative boundaries

02 CONVERTING RASTER DATA TO POLYLINES

- A polyline is created from each cell in the input raster dataset containing linear features.
- The polyline passes through the center of each cell.
- NoData cells do not become features in the output.

RASTER INPUT → POLYLINE OUTPUT

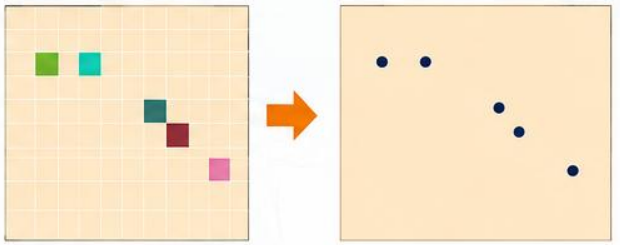


USE CASE
Road networks, rivers, pipelines, contour lines

03 CONVERTING RASTER DATA TO POINTS

- Each cell in the input raster containing point features converts to a point.
- Each new point is positioned at the center of the cell.
- NoData cells do not convert to points.

RASTER INPUT → POINT OUTPUT



USE CASE
Sampling locations, well locations, facility sites

GENERAL CONVERSION WORKFLOW

1. **INPUT DATA** — Raster dataset in GIS
2. **CONVERSION PROCESS** — Apply rules to convert data
3. **OUTPUT DATA** — Vector features (Polygon/Polyline/Point)
4. **VALIDATION** — Check accuracy and quality
5. **USAGE** — Ready for GIS analysis and decision making

WHY IT MATTERS?
Data conversion enables data integration, enhances analysis capabilities, and supports better spatial decisions.

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SPATIAL ANALYSIS

Understanding Where, Why and What Matters

1. WHAT IS SPATIAL ANALYSIS?

Spatial analysis is a **process** for looking at **spatial (geographic) patterns** in our data and at **relationships** between features. (Andy Mitchell, 1999).

OTHER DEFINITION

"The quantitative study of phenomena that are located in space exploiting their spatial component or location in space." (Bailey & Gatrell, 1995).

2. OBJECTIVE OF SPATIAL ANALYSIS

The principal objective of spatial analysis is:

- To transform and combine spatial & non-spatial data from diverse sources/disciplines into useful information.
- To improve our understanding about phenomena.
- To support and improve our decision making.

3. SPATIAL & NON-SPATIAL ANALYSIS

SPATIAL ANALYSIS

A A method of analysis is spatial if the results depend on the locations of the objects being analyzed.

- Move the objects, and the results change.
  - e.g. move the people, and the Malaysian Center of Population moves.

EXAMPLE

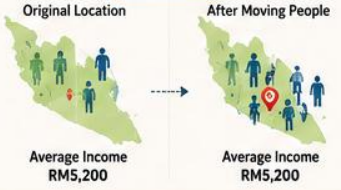


NON-SPATIAL ANALYSIS

A method of analysis is non-spatial if the results does not depend on the locations of the objects being analyzed.

- Most statistical techniques are invariant under changes of location.
  - e.g. move the people, and average income does not change.

EXAMPLE



SPATIAL ANALYSIS IN ACTION: ASEAN REAL-WORLD SCENARIOS

DISASTER RISK MANAGEMENT

Identify flood-prone areas in Indonesia by analyzing rainfall, elevation, land use and river networks.



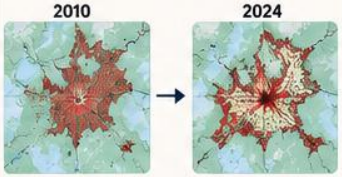
DENGUE HOTSPOT IDENTIFICATION

Detect dengue clusters in Thailand to target vector control and public awareness programs.



URBAN GROWTH MONITORING

Monitor urban expansion in Vietnam using satellite imagery and land cover change analysis.



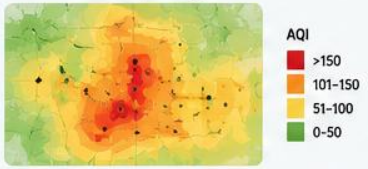
MARINE SPATIAL PLANNING

Plan sustainable marine zones in the Philippines by analyzing ecosystems, fisheries and shipping routes.



AIR QUALITY ASSESSMENT

Analyze air pollution distribution in Kuala Lumpur and surrounding areas to support policy and health planning.



WHY SPATIAL ANALYSIS MATTERS?

- Reveals patterns and trends in geographical space
- Supports evidence-based planning and policy making
- Helps allocate resources more effectively
- Enhances risk reduction and resilience
- Promotes sustainable development across ASEAN

WHY SPATIAL ANALYSIS IS IMPORTANT?



The cost of input to a GIS is high, and can only be justified by the benefits of analysis, modeling or decision-making performed with the data.

HIGH INPUT COSTS

- Satellite Imagery
- Field Survey & GPS Data
- Data Management
- Expertise & Software
- Time & Processing

HIGH VALUE BENEFITS

- Better understanding of spatial patterns and relationships
- Accurate modeling and prediction
- Stronger decision-making for better outcomes
- Efficient resource use & sustainable development

SPATIAL ANALYSIS – SUCCESSIVE LEVELS OF COMPLEXITY

Basic → Advanced

1. ELEMENTARY SPATIAL ANALYSIS (CLASSIC GIS CAPABILITIES)

Spatial queries & measurement, buffering, map layer overlay, network analysis, surface analysis etc.

EXAMPLES

- Find hospitals within 5 km
- Overlay land use with flood zones
- Shortest route for delivery
- Slope analysis for landslide risk

2. SPATIAL STATISTICAL ANALYSIS (HYPOTHESIS TESTING)

Are data "to be expected" or are they "unexpected" relative to some statistical model, usually of a random process.

EXAMPLES

- Detect spatial clusters of dengue cases in Thailand
- Identify areas with unusually high air pollution in Jakarta
- Test if deforestation patterns are random in Borneo

3. SPATIAL MODELING (PREDICTION)

Constructing models (of processes) to predict spatial outcomes (patterns).

What if analyses

EXAMPLES

- Predict future flood extent in the Mekong Basin
- Model urban growth in Ho Chi Minh City by 2035
- What if sea level rises by 1m along ASEAN coasts?

SPATIAL ANALYSIS IN ACTION: REAL-WORLD SCENARIOS IN ASEAN

FLOOD RISK MANAGEMENT



Identify flood-prone areas in Indonesia using rainfall, DEM, land use and river networks.

INDONESIA

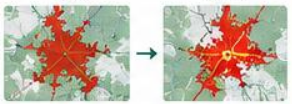
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THAILAND

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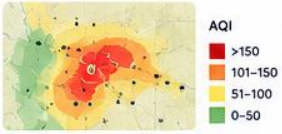
MARINE SPATIAL PLANNING



Plan sustainable marine zones in the Philippines for fisheries, conservation and shipping.

PHILIPPINES

AIR QUALITY ASSESSMENT



Analyze air pollution in Kuala Lumpur and surrounding areas to support policy and health planning.

MALAYSIA

DISASTER RESPONSE & EVACUATION PLANNING



Determine safe evacuation routes and shelter locations during typhoons in the Philippines.

PHILIPPINES

THE BIGGER PICTURE

- Reveals patterns and trends in geographical space
- Supports evidence-based planning and policy making
- Helps allocate resources more effectively
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- Promotes sustainable development across ASEAN



KNAMUKM

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THE SPATIAL ANALYSIS PROCESS

Turning Data into Location Intelligence

WHAT IS SPATIAL ANALYSIS?

Spatial analysis is a process for looking at **geographic patterns** in our data and at **relationships** between features.

METHODS WE USE CAN BE:

- Very simple – just by making a map
- More complex – involving models that mimic the real world by combining many data layers

WHY IS IT IMPORTANT?

The **cost** of input to a GIS is high, and can only be justified by the **benefits** of analysis, modeling or decision-making performed with the data.

- Better understanding of patterns & relationships
- Accurate modeling & prediction
- Stronger decision-making for better outcomes
- Efficient resource use & sustainable development

THE SPATIAL ANALYSIS PROCESS: 5 KEY STEPS

1 DEFINE OBJECTIVES & CRITERIA

- Frame your question and define criteria
- Specify how and who will use analysis
- Determine geographic boundaries
- Select appropriate map scale (resolution, precision, accuracy)

2 ASSEMBLE DATA

- Acquire existing data
- Understand data & metadata
- Create new data (digitize/field work)
- Check for errors & omissions
- Licensing & data sharing agreements

Satellite Imagery | Field Survey & GPS | Open Data Portals | Reports & Statistics | Sensors & IoT

3 PROCESS DATA

- Import/convert files
- Define coordinate system & match projections (e.g. Cassini Johor, RSO, RSO-GDM2000, WGS84, Lat-Long)
- Extract/modify features
- Update attribute information
- Generate symbology & explore data
- Document metadata

4 ANALYZE DATA

- Select by location or by attribute
- Create buffers, overlay layers
- Do calculations, apply complex models or spatial statistics
- Use multiple methods to get the information we need

5 PRESENT INFORMATION

- Interpret the output
- Decide if information is valid or if we should revisit steps
- Decide what to include in our map and how to group values
- Design map layout
- Create graphs & charts
- Export tables
- Write reports
- Online GIS & dashboards

TRANSFORMING DATA INTO ACTIONABLE INFORMATION

METHODS CAN BE AS SIMPLE OR AS COMPLEX AS NEEDED

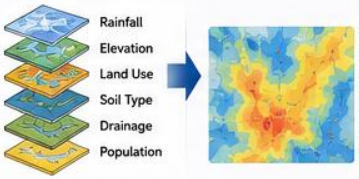
SIMPLE EXAMPLE

Create a map of schools within 5 km of a river.



COMPLEX EXAMPLE

Model future flood risk by combining rainfall, elevation, land use, soil, drainage, and population data.



SPATIAL ANALYSIS IN ACTION: REAL-WORLD SCENARIOS IN ASEAN

FLOOD RISK MANAGEMENT

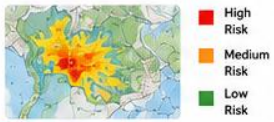
Identify flood-prone areas in Indonesia using rainfall, DEM, land use and river networks.



INDONESIA

DENGUE HOTSPOT IDENTIFICATION

Detect dengue clusters in Thailand to target vector control and public awareness programs.



THAILAND

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Monitor urban expansion in Vietnam using satellite imagery and land cover change analysis.



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GIS ANALYTICAL FUNCTIONS



Powerful Tools for Smarter Decisions

More than hundreds of functions

Can be classified into a number of categories

CATEGORIES OF GIS ANALYTICAL FUNCTIONS

Data Explorations
Explore, summarize and visualize your data to discover patterns and trends.

Vector Data Analysis
Analyze points, lines and polygons such as overlay, proximity and topology.

Raster Data Analysis
Analyze raster data like imagery and grids for classification, reclassification and overlay.

Terrain Mapping & Analysis
Analyze elevation, slope, aspect and terrain features.
- Viewsheds & Watersheds

Spatial Interpolation
Estimate values at unsampled locations using spatial interpolation techniques.

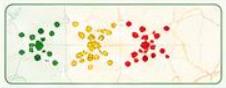
Geocoding & Dynamic Segmentation
Convert addresses to locations and segment customers or areas dynamically.

Path Analysis & Network Applications
Find the best routes, solve location-allocation problems and analyze networks.

Spatial Modeling
Build and use models to predict, simulate and support decision-making.

WHAT THESE FUNCTIONS CAN DO

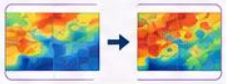
Data Explorations
- Summarize attribute data
- Find outliers and hotspots



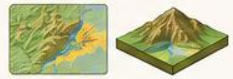
Vector Data Analysis
- Overlay multiple layers
- Find nearby features



Raster Data Analysis
- Classification, reclassification
- Change detection



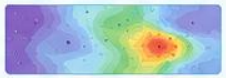
Terrain Mapping & Analysis
- Slope, aspect, elevation
- 3D visualization



Viewsheds & Watersheds
- Identify areas visible from a point
- Delineate drainage basins



Spatial Interpolation
- IDW, Kriging, Spline
- Create surfaces from sampled data



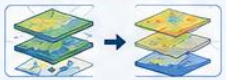
Geocoding & Dynamic Segmentation
- Address to location conversion
- Group customers by proximity or value



Path Analysis & Network Applications
- Shortest path, service area
- Location-allocation, routing

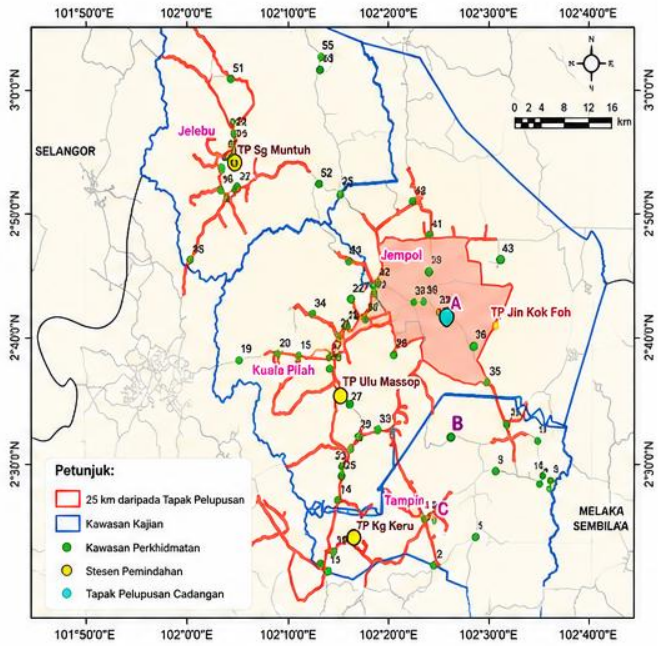


Spatial Modeling
- Predict future scenarios
- "What-if" analysis



EXAMPLE: OPTIMUM WASTE DISPOSAL SITE ANALYSIS

Optimum area can be reach solid waste disposal area (Sanitary) and transfer station within 25 km buffering area



REAL-WORLD APPLICATION SCENARIOS IN ASEAN

FLOOD RISK MANAGEMENT
Identify flood-prone areas in Indonesia using rainfall, DEM, land use and river networks.



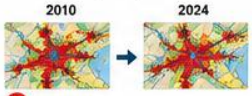
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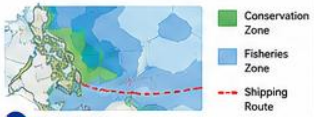
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MALAYSIA

DISASTER RESPONSE & EVACUATION PLANNING
Determine safe evacuation routes and shelter locations during typhoons in the Philippines.



PHILIPPINES

WHY GIS ANALYTICAL FUNCTIONS MATTER

- Reveals patterns and trends in geographical data
- Supports evidence-based planning and policy making
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GIS ANALYTICAL FUNCTIONS

Powerful Tools for Location Intelligence



More than **hundreds** of functions

Can be classified into a number of **categories**

CATEGORIES OF GIS ANALYTICAL FUNCTIONS

- Data Explorations
- Vector Data Analysis
- Raster Data Analysis
- Terrain Mapping & Analysis (Viewsheds & Watersheds)
- Spatial Interpolation
- Geocoding & Dynamic Segmentation
- Path Analysis & Network Applications
- Spatial Modeling

TWO POWERFUL GROUPS OF ANALYSIS

1 SPATIAL & NON SPATIAL ANALYSIS

Analyze and compare spatial relationships

- Draw conclusions
- Create new datasets



EXAMPLE SCENARIOS

- Compare air quality with hospital locations in Kuala Lumpur, Malaysia
- Analyze tourism hotspots vs. cultural heritage sites in Vietnam
- Compare land use change with economic growth in Indonesia
- Assess school accessibility vs. population distribution in the Philippines

2 ARC TOOLBOX

- Buffer, Multi-ring buffers
- Overlays (Intersect, Union)



EXAMPLE SCENARIOS

- Create 5 km buffer around airports in Bangkok to assess noise impact (Thailand)
- Multi-ring buffer for tsunami evacuation planning in Aceh (Indonesia)
- Overlay land parcels with flood zones in Cambodia (Intersect)
- Union overlay of protected areas and forest reserves in Laos

WHY THESE FUNCTIONS MATTER

- Reveal hidden patterns and relationships
- Support evidence-based planning and policy making
- Save time and resources with accurate analysis
- Help build resilient and sustainable communities across ASEAN

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ARC TOOLBOX IN ACTION: BUFFER & OVERLAY TOOLS

1 BUFFER

Draws a uniform area around a feature.
Represents a critical zone or zone of important.

KEY POINTS

- Features inside the buffer zone have different status from those outside.
- Example: Parcels inside a flood zone may have different insurance status.

HOW IT WORKS



Input Feature (Point/Line/Polygon) → Buffer Distance (e.g. 1 km) → Output Buffer Zone

EXAMPLE SCENARIOS IN ASEAN



Noise buffer around airports in Changi, Singapore

Floodplain buffer along rivers in Manila, Philippines

Setback buffer around industrial areas in Jakarta, Indonesia

2 MULTI-RING BUFFERS

Creates multiple buffer polygons around each feature.
Ascending distance polygons created.

HOW IT WORKS



EXAMPLE SCENARIOS IN ASEAN



Service area rings around hospitals in Hanoi, Vietnam

Multi-ring coverage around telecom towers in Kuala Lumpur, Malaysia

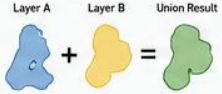
Accessibility rings around schools in Vientiane, Laos

USEFUL FOR Proximity analysis · Service area planning · Impact assessment

3 OVERLAYS (UNION & INTERSECT)

UNION OVERLAY

Combines layers, keeping all areas and merging attributes.



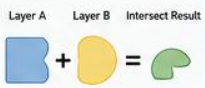
EXAMPLE IN ASEAN



What is the soil type and land cover for any given area in Cambodia?

INTERSECT OVERLAY

Keeps only the areas that overlap between layers.



EXAMPLE IN ASEAN



Tag county buildings with the soil type in which they are located.

OVERLAY OPERATIONS

- UNION = All areas from both layers
- INTERSECT = Only overlapping areas

COMMON USES IN ASEAN

- Land use planning
- Disaster risk mapping
- Environmental monitoring
- Infrastructure planning

Tip: Choose the right tool based on your question. → Need a zone around something? → Buffer → Need graduated distances? → Multi-Ring Buffer → Need to combine layers? → Overlay

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OVERLAY TOOLS

Combine Layers. Reveal Insights. Make Better Decisions.

Overlay tools help you identify spatial relationships by combining features from two or more layers.

COMMON USES IN ASEAN

- Land use & environment planning
- Disaster risk assessment
- Infrastructure & utilities planning
- Resource management

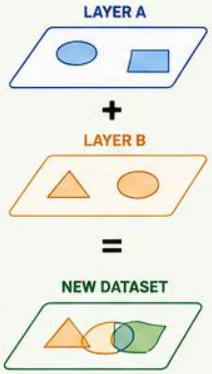
OVERLAYS OVERVIEW

Overlays identify overlaps between features in two layers and create a new dataset with new features.

Identify Overlaps
Find where features from different layers coincide.

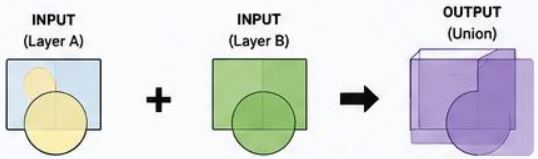
Create New Dataset
Generate new features based on overlaps.

Support Decisions
Use results for planning, policy, and analysis.



1 UNION OVERLAY

Combines layers, keeping all areas and merging attributes.



EXAMPLE SCENARIOS IN ASEAN

Soil Type + Land Cover



What is the soil type and land cover for any area in Thalland?

Protected Areas + Watersheds



Which protected areas fall within which watersheds in Laos?

Administrative Boundary + Infrastructure

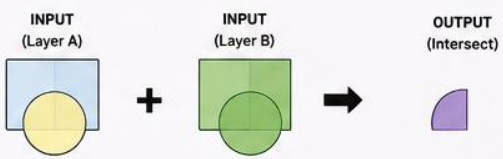


What roads and facilities fall within each district in Indonesia?

Best for combining all information from both layers.

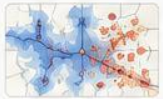
2 INTERSECT OVERLAY

Keeps only the areas that overlap between layers.



EXAMPLE SCENARIOS IN ASEAN

Flood Zone + Population



Which population lives within flood-prone areas in the Philippines?

Mangrove Area + Shrimp Farms



Which shrimp farms are located within mangrove zones in Vietnam?

Forest Cover + Conservation Areas



Which conservation areas are covered by forest in Malaysia?

Best for extracting only the common (overlapping) areas.

1A UNION OVERLAY DIALOG BOX (Arc Toolbox/Overlay)



- **Input** = Layers to union
- **Output** = Title and location of new dataset
- **Join Attributes** = Which attributes to be transferred
- **XY Tolerance** = Minimum distance separating features
- **Gaps Allowed** = Areas in output feature that are completely enclosed by other polygons



EXAMPLE (ASEAN)

Combine land cover and soil type to produce a comprehensive map for agricultural suitability in Cambodia.

2A INTERSECT OVERLAY DIALOG BOX (Arc Toolbox/Overlay)



- **Input** = Layers to intersect
- **Output** = Title and location of new dataset
- **Join Attributes** = Which attributes to be transferred
- **XY Tolerance** = Minimum distance separating features
- **Output Type** = Same as input, line, or point



EXAMPLE (ASEAN)

Find the intersection of typhoon track buffer and critical infrastructure to identify at-risk assets in the Philippines.

TIPS FOR BETTER OVERLAYS

- Use clean, accurate, and projected data.
- Ensure layers are in the same coordinate system.
- Set appropriate XY Tolerance for precision.
- Understand your data before combining.

KEY TAKEAWAY

Overlay tools turn multiple layers into meaningful information that supports better decisions for people and the environment.

SUMMARY

Spatial analysis uses location to understand patterns, relationships, and trends to support better decisions.



1. What is Spatial Analysis?

The process of examining data based on location to identify patterns, relationships, and trends.

2. Difference Between Spatial & Non-Spatial Analysis

Spatial analysis considers "where" things are; non-spatial analysis looks at data without location context.

3. Spatial Analysis Process

Define objectives ▶ Assemble data ▶ Process data ▶ Analyze data ▶ Present information

4. Coordinate Data Conversion

Converting coordinates ensures data from different sources aligns in the same reference system for accurate analysis.

REAL-WORLD SCENARIOS IN ASEAN

Flood Risk Mapping
Identify flood-prone areas to support early warning and community safety.



Philippines

Mangrove & Coastal Conservation
Monitor mangrove change and protect coastal ecosystems.



Indonesia

Infrastructure & Urban Planning
Find suitable locations for roads, utilities, and public facilities.



Vietnam

Agricultural Suitability
Analyze soil, slope & rainfall to improve crop productivity.



Thailand

Disaster Response & Relief Logistics
Plan routes and resource distribution during emergencies.



Malaysia

Spatial analysis turns location-based data into meaningful insights that help build safer, smarter, and more sustainable communities across ASEAN.

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ArcGIS Desktop – Data Conversion in Production Mapping
Guide to coordinate and data conversion for accurate mapping.
https://desktop.arcgis.com/en/arcmap/latest/extensions/production-mapping/data-conversion-in-production-mapping.htm

GIS Geography – Data Conversion
Simple explanations and examples of data & coordinate conversion.
https://gisgeography.com/tag/data-conversion/

WHY IT MATTERS FOR ASEAN

- Better disaster preparedness
- Sustainable natural resource management
- Smarter urban & rural planning
- Data-driven policy & decision-making
- Stronger regional collaboration

Accurate location. Better insights. Stronger ASEAN.